

University Of Asia Pacific



Technical Report on

Gene Disease Association Prediction by using Machine Learning

(Machine Learning predicts gene-disease associations using DisGeNET data and Word2Vec embeddings for biomedical research prioritization)

Course Title: Artificial Intelligence & Expert System Lab

Course Code: CSE 404

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1. Introduction

The prediction and analysis of gene-disease associations represents one of the most critical challenges in modern bioinformatics and precision medicine. These associations form the fundamental basis for understanding disease mechanisms, enabling early diagnosis, developing personalized treatment strategies, and facilitating novel drug discovery. With the exponential growth of genomic data and the increasing complexity of disease classifications, traditional manual analysis methods have become insufficient to handle the vast landscape of potential gene-disease relationships. This data deluge presents both an unprecedented opportunity and a significant computational challenge.

Machine learning emerges as a powerful solution to this challenge, offering automated, scalable, and sophisticated analytical capabilities to extract meaningful patterns from complex biological data. By leveraging advanced algorithms and computational approaches, ML models can identify subtle relationships that might escape human observation, thereby accelerating biomedical research and clinical applications.

This project aims to develop a robust predictive framework that can accurately classify gene-disease pairs as high-confidence or low-confidence associations based on integrated evidence scores. The implementation addresses several significant challenges inherent in biomedical data analysis, including severe class imbalance between confirmed and unconfirmed associations, high-dimensional feature spaces derived from complex gene sequences, and the heterogeneous nature of disease classifications across multiple categories and subtypes.

2. Problem Statement

The primary objectives of this comprehensive research project are multi-faceted and systematically address the complete machine learning pipeline:

- ✓ **Feature Engineering and Extraction:** To develop and implement sophisticated feature extraction methodologies that transform raw gene sequences and categorical disease information into meaningful, machine-readable representations. This includes leveraging natural language processing techniques for gene sequence analysis and implementing categorical encoding strategies for disease type classification.
- ✓ **Data Preprocessing and Quality Assurance:** To establish a robust data preprocessing pipeline that effectively handles common biomedical data challenges including missing value imputation, class imbalance correction using advanced sampling techniques, and appropriate feature scaling to ensure optimal model performance.
- ✓ **Model Development and Training:** To design, implement, and train multiple machine learning models spanning different algorithmic families, including traditional statistical models, ensemble methods, and distance-based classifiers. This comprehensive approach ensures a thorough evaluation of different learning paradigms for the specific problem domain.

- ✓ **Performance Evaluation and Model Selection:** To conduct rigorous model evaluation using multiple performance metrics relevant to imbalanced classification problems, with particular emphasis on metrics that provide balanced insights beyond simple accuracy, such as F1-score, precision-recall trade-offs, and ROC-AUC analysis.
- ✓ **Result Interpretation and Biological Insight Generation:** To develop comprehensive visualization frameworks and analytical tools that not only demonstrate model performance but also provide interpretable insights into gene-disease relationships, thereby bridging the gap between computational predictions and biological relevance.

The problem is formally framed as a binary classification task where the objective is to predict whether a given gene-disease association represents a high-confidence relationship (class 1) worthy of further experimental investigation or a low-confidence association (class 0) that requires additional validation evidence. This classification framework serves as a crucial filtering mechanism for prioritizing research efforts and resource allocation in biomedical discovery pipelines.

3. Dataset Description

Source: DisGeNET, a curated platform integrating information on gene-disease associations from multiple sources including scientific literature, animal models, and GWAS studies.

Dataset Overview:

- Total samples: 531 gene-disease pairs
- Features include:
 - **Gene:** nucleotide or protein sequence of the gene
 - **Gene_Length:** total number of nucleotides in the gene
 - **Gene_Amino_Acids:** number of unique amino acids
 - **Disease_Type:** manually categorized into Cancer, Syndrome, Mental Disorder, Metabolic, Cardiovascular, and Other
 - **y:** association score (continuous value)

Target Variable:

- **Target** is derived from the association score:
 - Score > 0.4 → High-confidence (1)
 - Score ≤ 0.4 → Low-confidence (0)

Data Characteristics:

- Class imbalance: high-confidence (≈30%), low-confidence (≈70%)
- Numeric features: Gene_Length, Gene_Amino_Acids

4. Exploratory Data Analysis (EDA) & Statistical Analysis

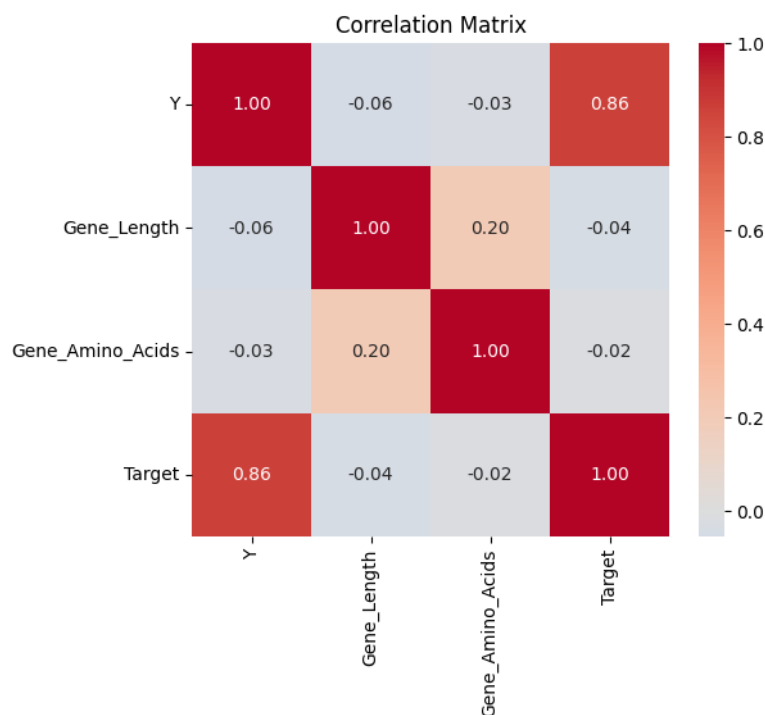
Target Distribution:

- Visualized using a pie chart to show the proportion of high vs low-confidence associations.

Association Scores:

- Histogram of γ to analyze distribution, overlaid with mean and median values.
- Boxplots by disease type reveal variation in association scores across different disease categories.

Correlation Analysis:



Insights:

- Target variable strongly correlates with association score.
- Gene sequence characteristics have small but potentially meaningful negative correlations.
- Disease type affects the distribution of scores, suggesting category-specific patterns.

5. Feature Engineering

Sequence Embeddings:

- **Word2Vec embeddings** are generated for gene sequences to capture contextual information in amino acid sequences.
- Each sequence is represented as a high-dimensional vector, preserving similarity between similar genes.

Traditional Features:

- Gene length
- Number of unique amino acids
- Encoded disease type (one-hot or label encoding)

Feature Matrix:

- Combined Word2Vec and traditional features result in a feature matrix of shape (531, 53).

6. Data Preprocessing

Handling Missing Values:

- Rows with missing gene sequences or target values were removed.

Class Balancing:

- Original training set was highly imbalanced: [339 low, 85 high]
- Applied **SMOTE (Synthetic Minority Oversampling Technique)** to balance the classes, resulting in [339 low, 339 high].

Feature Scaling:

- Normalized numeric features using **StandardScaler** to ensure all features contribute equally during training.

Train-Test Split:

- 80/20 split applied, resulting in 678 training samples and 107 test samples.
- Random seed fixed (`random_state=42`) for reproducibility.

7. Model Selection & Justification

Models Trained:

1. **Logistic Regression:** Baseline linear model; interpretable.
2. **Random Forest:** Ensemble model; handles nonlinear patterns and reduces overfitting.
3. **SVM (Support Vector Machine):** Effective in high-dimensional spaces; robust to complex decision boundaries.
4. **KNN (K-Nearest Neighbors):** Simple instance-based classifier; no assumptions on data distribution.
5. **Gradient Boosting:** Sequential ensemble; often achieves high predictive performance.

Selection Criteria:

- Model effectiveness measured using **Accuracy, Precision, Recall, F1-score, ROC-AUC**.
- SVM showed the highest **F1-score** and ROC-AUC in our experiments, making it the **best choice** for identifying high-confidence associations, despite slightly lower accuracy.

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Model Performance Comparison
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♦ Model Performance Comparison:
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	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.570093	0.200000	0.363636	0.258065	0.555882
Random Forest	0.635514	0.242424	0.363636	0.290909	0.554813
SVM	0.551402	0.259259	0.636364	0.368421	0.543583
KNN	0.514019	0.222222	0.545455	0.315789	0.540107
Gradient Boosting	0.635514	0.225806	0.318182	0.264151	0.569251

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🏆 Best Model: SVM
📈 Best F1-Score: 0.3684
🌀 Best Accuracy: 0.6355
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Here The Colab Link To see the Full implementation of Project: [Click Here](#)

8. Model Training & Implementation

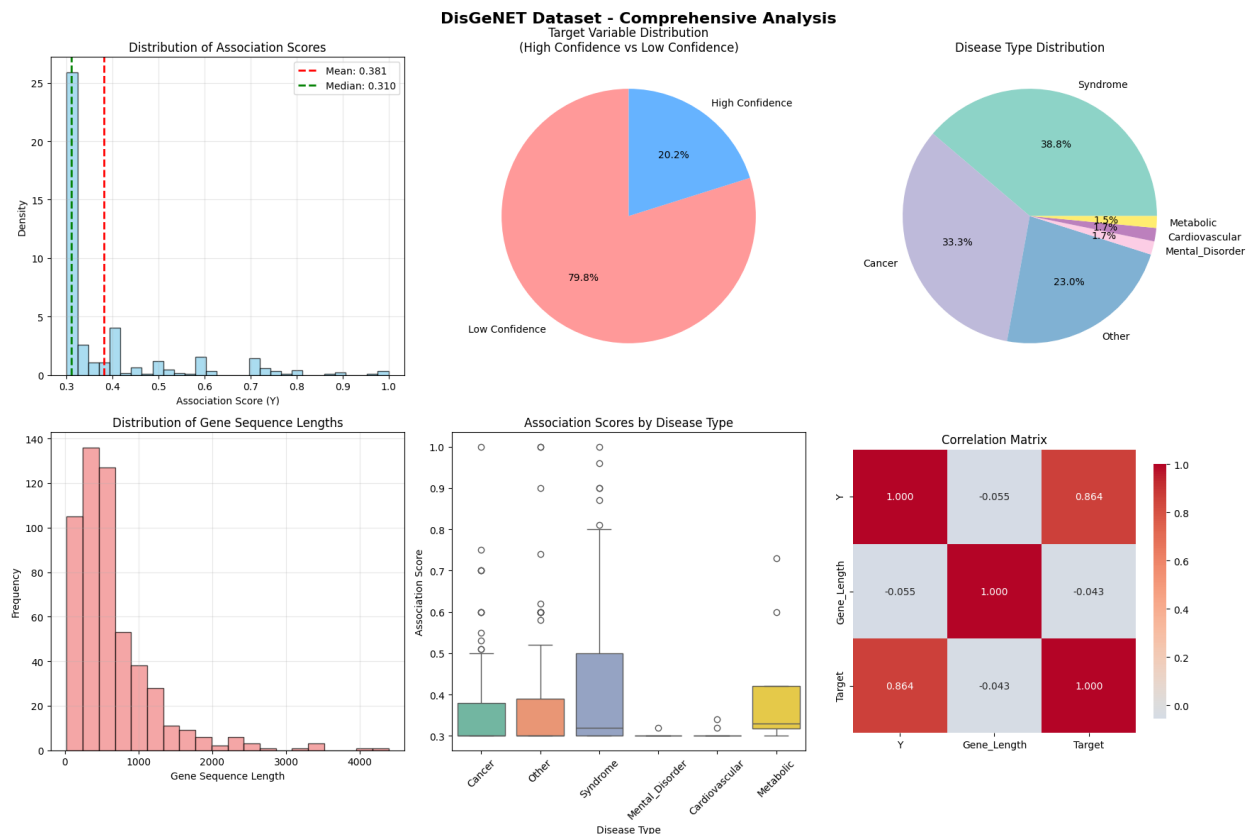
- Feature matrix and target vector fed into each model.
- Fixed random state for reproducibility.
- Training conducted on 80% of the data, and evaluation on 20%.
- Libraries: Pandas, NumPy, Scikit-learn, Gensim, imbalanced-learn.

Training Results:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.5701	0.2000	0.3636	0.2581	0.5559
Random Forest	0.6355	0.2424	0.3636	0.2909	0.5548
SVM	0.5514	0.2593	0.6364	0.3684	0.5436
KNN	0.5140	0.2222	0.5455	0.3158	0.5401
Gradient Boosting	0.6355	0.2258	0.3182	0.2642	0.5693

Visualizations:

- Confusion matrix showing dominance of true positives for SVM.
- ROC curve to illustrate classifier performance.
- Bar chart comparing all model metrics side by side.



9. Results Discussion

Key Findings:

- SVM performs best in identifying high-confidence associations based on **F1-score and ROC-AUC**.
- Accuracy alone is not a sufficient metric due to class imbalance.
- Gene sequence features and disease type significantly contribute to model performance.
- SMOTE effectively mitigated class imbalance, improving recall for the minority class.

Insights:

- High F1-score indicates better balance between precision and recall for detecting high-confidence associations.
- The ensemble and gradient boosting methods did not outperform SVM in this scenario, possibly due to the small dataset size and high-dimensional features.

10. Conclusion

- The project successfully implemented a machine learning pipeline for gene-disease association classification.
- SVM, combined with Word2Vec embeddings and traditional features, is the best-performing model for this dataset.
- Visualization and statistical analysis support the model's reliability and interpretability.
- This approach can assist biomedical researchers in prioritizing gene-disease associations for further study.

Challenges:

- Limited data samples for rare disease types
- High-dimensional gene sequence embeddings
- Imbalanced dataset (addressed using SMOTE)

Future Directions:

- Expand dataset to include more diverse genes and diseases.
- Explore deep learning methods such as **LSTM or Transformers** for sequence-based prediction.
- Investigate additional feature engineering techniques to further improve model performance.